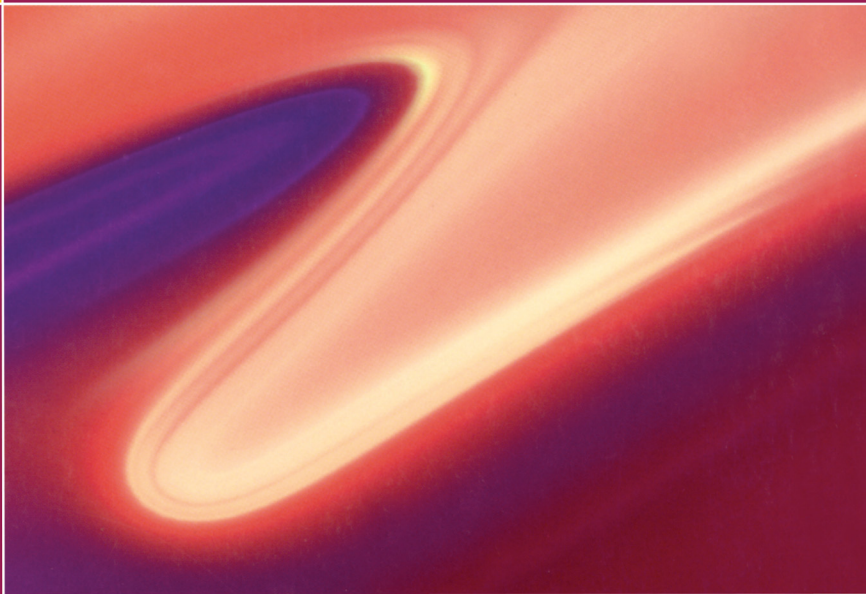


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Plasma Physics via Computer Simulation

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Foreword

The complex nature of the problems encountered in plasma physics has motivated considerable interest in computer simulation, which has played an essential role in the development of plasma theory. In addition, computer simulation is also becoming an efficient design tool to provide accurate performance predictions in plasma physics applications to fusion reactors and other devices, which are now entering the engineering phase.

Computer simulation of plasmas comprises two general areas based on kinetic and fluid descriptions, as shown in Figure a. While fluid simulation proceeds by solving numerically the magnetohydrodynamic (MHD) equations of a plasma, assuming approximate transport coefficients, kinetic simulation considers more detailed models of the plasma involving particle interactions through the electromagnetic field. This is achieved either by solving numerically the plasma kinetic equations (e.g. Vlasov or Fokker-Planck equations) or by "particle" simulation, which simply computes the motions of a collection of charged particles, interacting with each other and with externally applied fields. The pioneering work of Dawson and others in

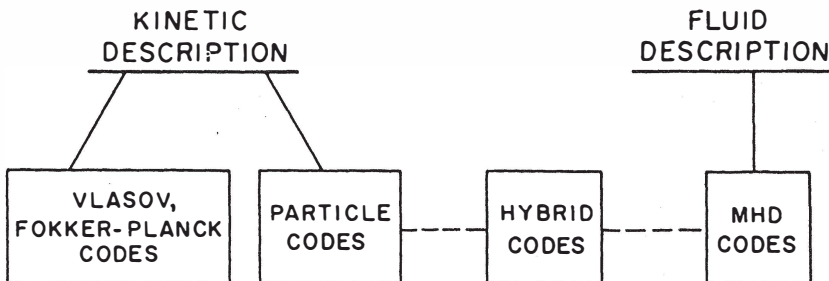


Figure a Classification of computer simulation models of plasmas.

the early 60's has shown that, when appropriate methods are used, relatively small systems of a few thousand particles can indeed simulate accurately the collective behavior of real plasmas. Since then, the development of new algorithms and the availability of more powerful computers has allowed particle simulation to progress from simple, one-dimensional, electrostatic problems to more complex and realistic situations, involving electromagnetic fields in multiple dimensions and up to 10^6 particles. Kinetic simulation has been particularly successful in dealing with basic physical problems in which the particle distributions deviate significantly from a local Maxwellian distribution, such as when wave-particle resonances, trapping, or stochastic heating occur. MHD simulation, on the other hand, has generally been applied to large-scale problems directly related to the behavior of experimental devices. However, this simple distinction between kinetic and MHD simulations is becoming more complex through the increasingly-common use of "hybrid" codes, in which for example, fluid and particle treatments are applied to different components of a given plasma, and through the introduction of particle-hydrodynamic codes, in which fluid equations are solved by particle methods. The recent development of implicit algorithms is also expected to allow application of particle, Vlasov or Fokker-Planck codes to long-time-scale transport problems, which have generally been treated by fluid simulation.

Both MHD and kinetic simulations, including particle simulation, are well-developed disciplines, which have become an integral part of plasma physics, and the need for textbooks and basic references in these areas has been felt for some time. "Plasma Physics via Computer Simulation," by C. K. Birdsall and A. B. Langdon, provides a clearly-written and competent answer to this need in the area of particle simulation. The book consists of three parts, Part One presenting elementary particle simulation methods, while Parts Two and Three deal with the fundamental numerical analysis problems of particle simulation, and with more advanced particle simulation techniques applicable to electromagnetic fields and to problems in several dimensions. The elementary description of Part One is supported by the electrostatic code ES1 and electromagnetic code EM1, and proceeds with a number of physically interesting projects. This is a very appropriate approach in a field which can best be learned by actual practice, and as implied in the title of the book, computer simulation can itself be a useful pedagogical tool. For example, phase-space plots from a particle simulation can be worth "a thousand equations" to illustrate nonlinear wave-particle interactions. Thus, this part of the book is not limited to specialists in computer simulations, but should be of interest to plasma physicists in general, and should provide a better understanding of the capabilities and limitations of these methods. Part Two and Three are addressed to readers who need a deeper knowledge of particle simulation. These parts provide a knowledgeable presentation of advanced subjects and guidance through the extensive literature of this field.

Ned Birdsall and Bruce Langdon are recognized authorities in plasma physics subjects associated with particle simulation. Both have made important contributions to the development and to the applications of particle simulation in plasma physics problems related to magnetic- and inertial-confinement. The book has evolved from class notes, compiled over a decade of teaching of the subject at Berkeley, which have already served a generation of students, many of whom are now well established in the field. The authors have also called on several contributors to present certain subjects, and the present work represents a comprehensive text on the fundamental aspects of particle simulations.

J. Denavit

Lawrence Livermore Laboratory

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Preface

Our book is on particle simulation of plasmas, aimed at developing insight into the essence of plasma behavior. Major current applications are to magnetically- and inertially-contained fusion plasmas. However, particle simulations are also being used to gain understanding of plasmas in space, and man-made plasmas such as occur in electron and ion guns, plasma propulsion and microwave devices, and in nuclear explosions. Our title notwithstanding, we make no pretense of covering all of plasma physics or all of computer simulation.

Plasma is the fourth state of matter, consisting of electrons, ions and neutral atoms, usually at temperatures above 10^4 degrees Kelvin. The sun and stars are plasmas; the earth's ionosphere, Van Allen belts, magnetosphere, etc., are all plasmas. Indeed, plasma makes up much of the known matter in the universe.

Plasma is the medium for magnetically or inertially-confined controlled thermonuclear fusion. A plasma of deuterium and tritium ions heated to a temperature of 10^8 degrees Kelvin undergoes thermonuclear burn, producing energetic helium ions and neutrons from fusion reactions. Such plasmas could be used as sources of heat which may be used to produce steam to run turbines to drive electric generators. Current studies of fusion plasmas include theory, experiment, and computation using large and fast computers. The latter is very extensive because it has produced results useful to theory and experiment and because current fusion plasma experiments cost considerable time (years) and money (hundreds of millions of dollars). Included in computation is plasma simulation; part is done using fluid models; part is done using many-particle models (meaning 10^3 to 10^6 particles) in order to obtain detailed kinetic behavior; part is done using hybrid models with both fluids and particles.

Plasma simulation using particles has grown from art to science in the past twenty years, and is now in world-wide use. The cost has diminished as computers have improved to the point where the simulation codes presented in the text may be run on small computers or run very quickly on large computers. Hence, particle simulation is used not only by the large laboratories, but is well within reach of small university groups. And, very ambitious programs may be run on large computers, as through the network connecting universities and laboratories to the National Magnetic Fusion Energy Computer Center at Livermore.

The text has four major Parts:

Part One, *Primer*. This part is intended for the novice who wishes to do plasma simulation with particles by actually simulating. The computer code ES1 [an electrostatic one dimensional (1d) code] is described, with a listing. EM1 (an electromagnetic 1d code) is also described, along with the numerical methods needed to do the computational physics. The Primer is useful for an introductory course, complete with problems, or for the beginning of a longer course. Projects are provided, with suggested initial values and selected results.

Part Two, *Theory*. This part provides most of the current theory on electrostatic particle simulation, helping to explain the effects of finite time steps and of the spatial grid on the plasma physics. Part Two provides the mathematical and physical foundations for the algorithms used in Part One, and their effects.

Part Three, *Practice*. This part covers more complicated simulations in two dimensions, both electrostatic and electromagnetic; it is close to current plasma research, and is intended for use by research people and by students.

Part Four, *Appendices*. These provide many of the details essential to running computer simulations.

These features (programs/projects/problems/theory) were developed for a course which has been taught over the past decade in Berkeley.

A result rewarding to us in doing plasma simulations has been the fun of obtaining thorough physical understanding of plasma behavior. There is considerable excitement in making a program produce good physics. We recommend that the projects be started during the first week of class so that students are doing simulations as the theory is being covered in lectures and reading. The quiz questions in Chapter 3 may be used to ensure reading and understanding of the program. Simulation is almost always the only *direct* experience students have with plasma oscillations, streaming instabilities and the kinetic behavior of warm plasmas; almost all of their other experiences (in theory and lab) are *indirect*. Our experience has been that those students willing to work through the projects in Part One and add the theoretical understanding of Part Two will have deepened their insight into plasma behavior, whether they continue in theory or simulation or experiment.

We quote from the introduction to "Elements of Color," by Johannes Itten: Learning from books and teachers is like travelling by carriage, so we

are told in the Veda. The thought goes on, "But the carriage will serve only when one is on the highroad. He who reaches the end of the highroad will leave the carriage and walk afoot." Doctrine and theory are best for weaker moments. In moments of strength, problems are solved intuitively, as if of themselves.

We have emphasized mathematical tools with which to construct algorithms with desired properties and analyze algorithms. Many of the algorithms were developed without use of such tools by people whose style and intuition leads to successful algorithms. Many useful codes have been assembled in an *ad hoc* manner and (often) work well, even though it may be impractical to calculate analytically the effects of finite Δx and Δt on the outcome.

This book gathers information which is valuable to simulators, some of which is scattered through published journals, and some of which is unpublished. Hence, it is intended also for reference use.

We are delighted to recommend "Computer Simulation Using Particles" by R. W. Hockney and J. W. Eastwood (1981, McGraw-Hill) as a complementary text. Where our emphasis is on plasma simulation, they extend the techniques developed primarily for plasmas to simulations of semiconductor devices, of gravitational problems, and of solids and liquids.

Charles K. (Ned) Birdsall
A. Bruce Langdon

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Birdsall thanks W. B. Bridges for our work with electron diode simulations begun in 1959, which in turn benefited from the pioneering work on electron device simulations in the 1950's of T. G. Mihran and S. P. Yu and of P. K. Tien. A. Hasegawa introduced me to 1½d models in 1962. J. A. Byers helped me on 2d Poisson solvers and linear weighting in 1964. T. Kamimura helped me with 2d and 3d gridded simulations in 1966 in Osaka, as did D. Fuss in 1967-1970, and N. Maron in 1972-75, at LLNL, Livermore. Collaboration with Langdon began in 1967 and has been most challenging and productive.

Langdon began plasma simulation with J. M. Dawson, who sets an exceptional example of the symbiosis of theory, simulation, and intuition. Much of the theoretical understanding of simulation methods and applications was done with, or stimulated by, Birdsall, who has catalyzed many successful projects and careers. Many collaborations have been instructive, especially the Astron simulations with J. Byers, J. Killeen, and others, and

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Our book originated as a set of class notes intended for use by graduate students who were learning to simulate using ES1. The first set was written about 1973; the second set, then in two parts (Primer and Theory) was finished in 1975; the third set, with the theory part rewritten and Practice and Appendices added, was completed in 1978. The current text thus contains sections written over most a decade during which our secretaries struggled in typing from pretty rough notes, namely Paula Bjork, Pamela Humphrey, and Michael Hoagland in Berkeley and Jill Dickinson in Reading, to whom we are most grateful.

The production team for putting the final version into camera-ready form during 1980-1984, was lead by Douglas W. Potter, who developed the macros and was responsible for the final photocopies. H. Stephen Au-Yeung and Carolyn Overhoff typed much of the book and the corrections. Thomas King and Fiona E. O'Neill, did all of the drawings, usually starting from rough hand-drawn sketches. Ginger Pletcher located references, handled correspondence and other tasks. Thomas L. Crystal coordinated the production during the last three years and was responsible for much of the typing and final corrections in Chapters 12-16. Our team performed admirably

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We acknowledge, with thanks, permission from authors and editors and publishers to draw on material published in journals and in books and modified to conform to our text style. A list of publishers follows:

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Conferences on Numerical Simulation of Plasmas (starting in 1967, with the tenth held in 1983)

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We have been most fortunate to work with patient and professional editors and associates at McGraw-Hill, notably B. J. Clark, David Damstra, Madelaine Eichberg, Diane Heiberg, and T. Michael Slaughter, who suffered through our pioneering task of producing camera-ready copy in Berkeley.

Charles K. (Ned) Birdsall
A. Bruce Langdon

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PART ONE

PRIMER

ONE DIMENSIONAL ELECTROSTATIC AND ELECTROMAGNETIC CODES

This part of the book is truly a primer for plasma simulation using particles and is intended for use by those with some knowledge of plasmas and some ability in numerical methods and programming. However, the reader with no prior plasma or numerical experience may still profit from this part, using additional texts on plasmas.

These seven chapters have been used in teaching particle simulation for roughly a decade. The lectures follow the chapters as written. The student homework, however, begins in the first week with assignment of the projects of Chapter 5. Hence, students are actively involved in running a one-dimensional electrostatic code from the first day of class.

Chapter 1 is introductory, and is intended to make the reader feel comfortable with using a few hundred to a few thousand particles to simulate a laboratory plasma of perhaps 10^{14} to 10^{24} particles.

Chapter 2 presents some details of the one-dimensional electrostatic program ES1 in terms of the differential and difference equations to be solved. The algorithms, initial values, and diagnostics are described briefly.

Chapter 3 presents ES1 in detail and includes the listings of most parts of the program. Some of the problems are used as classroom quizzes.

Chapter 4 goes into the numerical details of the particle mover, the meaning of particle shapes as inferred from weighting from particle to grid and grid to particle, and the Poisson equation solver.

Chapter 5 presents a number of introductory projects on plasma oscillations, waves, and instabilities for students to run using ES1. Details are presented on choice of initial conditions, development of linear analysis to check against, and approximate nonlinear analysis. These projects start with cold plasmas which are very easy and inexpensive to run and to understand. Successful completion of most or all of these projects is essential as an introduction to professional plasma particle simulation. This chapter (and Chapter 7) are the laboratory parts of the course.

Chapter 6 describes a one-dimensional electromagnetic program EM1 which uses Maxwell's equations for the transverse fields. Chapter 7 presents several projects for the student to run using EM1.

While Part One may stand alone for a course in particle simulation, teachers and students are encouraged to read ahead in Part Two for more complete development of the theory of particle simulation, especially for insight into some of the results they obtain in the projects.

WHY ATTEMPTING
TO DO PLASMA PHYSICS
VIA COMPUTER SIMULATION
USING PARTICLES
MAKES GOOD PHYSICAL SENSE

The idea of obtaining more or less valid plasma physics by using a computer to follow charged particles occurred to a number of people, notably John Dawson at Princeton and Oscar Buneman at Cambridge, in the late 1950's and early 1960's. After all, there had been simulations following particles of electron beams in vacuum tubes all during the 1950's, so why not simulate plasmas? It simply was not clear that taking the step from cold beams with charges all of one sign to thermal distributions with essentially equal density of charges of opposite signs and greatly different masses would be successful.

Why not?

In simulating an electron beam, it is reasonable to think that the computations would be valid if some small number of particles like 16 or 32 is used per wavelength. These particles are usually disks with diameter of the beam and are followed by the computer (like buttons on a string, except that they are tenuous and allowed to overtake each other) during their interaction with microwave circuits (say, resonant cavities or slow-wave structures) over some five or ten wavelengths. A total of 10 times 16 or 32 particles are used and followed for, say, 10 to 20 cycles (a few hundred time steps) from linear modulation through to nonlinear saturation. Once computers became easily

programmable, these simulations became rather straightforward. These simulations succeed with few particles because the field of one electron acts on a large fraction of the electrons; the collective interaction length is comparable to the dimensions of the electronic device.

In considering simulating *neutral* plasmas, one first turns to the early pages of numerous plasma texts, where the Debye length is introduced. This length is numerically related to the *plasma frequency* $\omega_p \propto (\text{density})^{1/2}$, by

$$\lambda_D \equiv \frac{v_{\text{thermal}}}{\omega_p} \propto \left(\frac{\text{temperature}}{\text{density}} \right)^{1/2} \quad (1)$$

This is the distance traveled by a particle at the thermal velocity in $1/2\pi$ of a plasma cycle, the shielding distance around a test charge and the scale length inside which particle-particle effects occur most strongly and outside of which collective effects dominate. We also read the assertion that "plasmas of interest are many λ_D across, $L \gg \lambda_D$." In order to have many particles within a collective interaction length, as in the electronic devices, we must have a much larger number of particles in simulating a neutral plasma.

(It is assumed that our readers will have experience with plasmas through lecture and/or laboratory courses and will turn to plasma texts for help when needed. Although some elementary statements or definitions are given in the text, not all of the plasma physics required for our text will be developed; our text is aimed at complementing plasma texts.)

It may seem worse. In the texts we often find a chart of density (from about $n = 10^6/\text{cm}^3$ to $n = 10^{22}/\text{cm}^3$) versus energy (from about 0.01 eV to 10^5 eV) with lines of constant N_D , the number of particles in a Debye cube,

$$N_D \equiv n\lambda_D^3 \quad (2)$$

Although very dense plasmas can have small N_D , we see that alkali vapor plasmas have $N_D \approx 10^2$, the earth's ionosphere has $N_D \approx 10^4$, and magnetically confined fusion experiment plasmas have $N_D \approx 10^6$! A literal simulation of the latter experiments is inconceivable in the foreseeable future.

Hence, we might be tempted to stop and say that simulation is hopeless, requiring many orders of magnitude more particles than can be handled by any existing computer.

If we succumb to this argument, then we miss the general character of much of plasma behavior. First, we are often going to be interested in the *collective behavior of collisionless plasmas at wavelengths longer than the Debye length*, $\lambda \gtrsim \lambda_D$. Second, we can obtain much useful information in one and two dimensions where we need not satisfy the three-dimensional requirement, say, $N_D = 10^6$, but only the one-dimensional $N_D = n\lambda_D \approx (10^6)^{1/3} = 10^2$ or the two-dimensional $N_D = n\lambda_D^2 \approx (10^6)^{1/2} = 10^3$. Third, we can alter the simulation so that we may use even smaller N_D but keep the essential plasma behavior. *The models we will use are intended to produce the essence of the plasma, without all of the details.*

Let us examine more closely the rough statements just made. To be sure, a collisionless laboratory plasma is characterized by $N_D \gg 1$ and $L \gg \lambda_D$. However, the physical behavior of a plasma is one of electrons and ions moving in their Coulomb fields with sufficient kinetic energy to inhibit recombination. Hence, another characterization of a plasma is that

$$\frac{\text{Thermal kinetic energy (KE)}}{\text{Microscopic potential energy (PE)}} \gg 1 \quad (3)$$

This ratio, for laboratory plasmas, is indeed N_D . However, the fundamental physics only requires $\text{KE} \gg \text{PE}$, which may be quite satisfactory at N_D as low as 10, or which may be achieved by means other than requiring $N_D \gg 1$, if necessary. The second characterization, collisionless, is also tied to N_D through

$$\frac{\nu}{\omega_p} \approx \frac{1}{N_D} \ln N_D \ll 1$$

and may likewise either be acceptable for $N_D = 10$ or achieved by means other than $N_D \gg 1$. The third description, $L \gg \lambda_D$, can be realized, in many problems, by use of periodic models which look at a slice of an infinite plasma, and use, for example, $L = 50\lambda_D$. This paragraph says that we may choose to simulate warm plasmas at small values of N_D or small L/λ_D . Examination of the physics of cold plasmas (e.g., Langmuir oscillations at $\omega = \pm\omega_p$) and of nonneutral plasmas (ion density not equal to electron density in zero order) also shows that simulation with particles is practical.

What are these "means other than $N_D \gg 1$ "? We use several in our simulations. First, we seldom keep track of interactions down to infinitesimal charge separations. In 3d (shorthand for 3 dimensions) and 2d, the impact parameters are always relatively large even for $N_D \approx 10$. In 1d, collisions are of a different nature; also, in 1d, making $N_D = n\lambda_D$ large is not a difficulty. If we use a *spatial grid* in order to simplify the calculation of the fields, then fields and forces at lengths less than a grid cell are not observable and may be considered to be smoothed away; we can do even more *smoothing* by such means as dropping forces at short wavelengths (large wave vectors, $k = 2\pi/\lambda$), by dropping charge densities, potentials, and fields at large values of k .

Second, we may deliberately alter the physics in order to achieve either $\text{KE} \gg \text{PE}$ or $\nu \ll \omega_p$, or both. For example, laboratory particles are generally considered to be point particles, with forces between particles separated by distance r given by $1/r^2$, $1/r^1$, or $1/r^0$ in 3d, 2d, and 1d. These Coulomb singularities in 3d and 2d (there are none in 1d) may be removed by using *finite-sized particles*, a notion employed long before plasma simulation; for example, see *Vlasov* (1950), who calls his particles *clouds*, as we do also. The important results here are that:

- (a) Finite-size particles occur naturally when we use a spatial grid.
- (b) As the finite-size particle radius R is made comparable or larger than the Debye length ($R \geq \lambda_D$), then the collision cross-section σ and collision frequency ν diminish rapidly relative to that of point particles, $R = 0$ in 2d and 3d.
- (c) There is considerable latitude for invention in terms of particle and force weightings relative to the spatial grid in order to achieve results desired in any one simulation problem (applicable in one but perhaps not in another).

Nonetheless, we simulate with far fewer particles than are in the laboratory plasma of interest. As a result, we usually live with relatively higher collision rates, *e.g.*, $\nu \ll \omega_p$, but not $\nu \ll \ll \omega_p$ (and altered collisional dependence on parameters), somewhat higher noise levels (our *superparticles* may have the same charge-to-mass ratio as in the lab but each q and m is much larger), and inability to examine all time and space scales (all frequencies ω and wave vectors k).

An alternative to simulation using particles is integration of the collisionless kinetic ("Vlasov") equation, which treats phase space as a continuum (which is also an approximation). This approach does avoid statistical errors present in particle simulation, and has been used successfully. Vlasov simulation has not so far proven to be as adaptable as particle simulation, especially in multidimensional problems, and untempermental, accurate, yet economical, representation of velocity space has been difficult in long-time simulations.

We simulate only over limited time and space and so can tolerate small errors. We even use mass ratios $m_{\text{ion}}/m_{\text{electron}}$ like 100 rather than 1836 (proton/electron) or sometimes freeze the motion of the neutralizing or background particles or put in a linearized susceptibility $\chi(\omega, k)$ say, for warm electrons. We find that the finite time and space gridding may itself produce waves, even instabilities, that are nonphysical and, hence, unwanted.

Indeed, we are accused of tomfoolery more than we deserve. We then simply admit to being in good company with the rest of plasma physics, with theorists and experimentalists who also have their kit bags of approximate (and occasionally inaccurate) tools.

The point is, within all three branches of plasma study (experiment, theory, computation), practitioners must exercise a great deal of care, enough to obtain the essence of the problem, but not so much as to inhibit achieving any result. We need only do well enough. This text is intended to provide insight into commonly used methods, to offer practice problems and projects and to give the reader a starting place for doing his own simulations.

Hockney and Eastwood (1981, chapter 1) provide similar and elegant reasoning for performing computer experiments using particle models as applied to plasma physics and to a variety of other physical areas.

OVERALL VIEW OF A ONE- DIMENSIONAL ELECTROSTATIC PROGRAM

2-1 INTRODUCTION

In this chapter, without becoming overly involved in the details of the various parts, we present a brief once-through of a particle simulation program. At the same time, we hope that the reader does not think that the whole program is too elementary; we will take up improvements, subtleties, and alternatives later.

Our procedure is to follow a widely-used program ES1, an ElectroStatic 1-dimensional program, complete with initial conditions and diagnostics to tell us about the physics, as well as check on the numerics. This is preparatory to running the program from tens to thousands of time steps on trial runs. We may find out that we were not as wise as we thought on the first run; hence, we will modify the program, improve the initial conditions, add some new diagnostics and try again. After a few such go-arounds as this, we may have what we started out after, the essence of the physics. Of course, along the way, we probably will find that we want more solutions to the dispersion equation for the waves that we were studying, or better estimates of the nonlinear behavior to be observed, or other information.

Our initial system for study has charged particles in both self and applied electrostatic fields and an applied magnetic field. This initial use of an electrostatic model follows the historical development of plasma particle

simulation, is perhaps the easiest of all models to understand, and also leads directly into fully electromagnetic models.

We do this in a spirit of complementing theory or experiment, in order to observe new phenomena, or to understand what has been predicted or observed in the laboratory. It is wise to remain coupled to both theory and experiment.

2-2 THE ELECTROSTATIC MODEL: GENERAL REMARKS

The model (Figure 2-2a) consists of charged particles moving about due to forces of their own and applied fields. The physics comes from two parts, the fields produced by the particles and the motion produced by the forces (or fields). The fields are calculated from *Maxwell's equations* by knowing the positions of all of the particles and their velocities; the forces on the particles are found using the electric and magnetic fields in the *Newton-Lorentz* equation of motion. One calculates the fields from the initial charge and current densities, then moves the particles (small distances) and recalculates the fields due to the particles at their new positions and velocities; this procedure is repeated for many time steps.

The difference from a laboratory plasma is that simulations proceed discontinuously in time *step by step*, using *digital* rather than analog computation. We must show care in developing *numerical methods* that provide sufficient *accuracy and stability* to make the simulations useful through many characteristic cycles of the plasma, whether these be plasma, cyclotron, or hybrid (or whatever) periods of the ions or electrons. We use a *temporal grid*

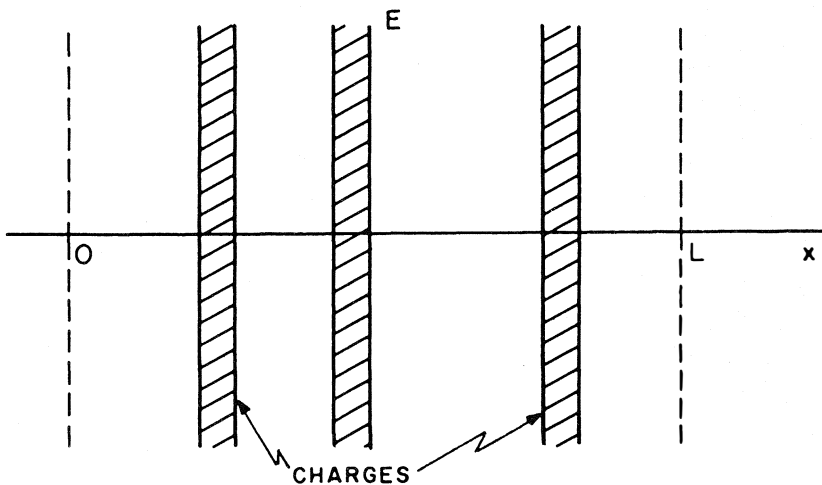


Figure 2-2a A one-dimensional model, consisting of many sheet charges, with self and applied electric fields E directed along the coordinate x . There are no variations in y or z .

which is sufficiently fine grained to follow the plasma with acceptable accuracy and stability.

A second difference is the use of a *spatial grid* on which the fields will be calculated. One might ask, why not use *Coulomb's law* directly (for forces between charges separated by distance r , where the force decays as $1/r^2$, $1/r^1$, $1/r^0$ in three, two, and one dimensions)? Consider the calculation of these forces in terms of both the numerical operations required and in terms of the actual physics; *i.e.*, are we interested in the details of close encounters among particles and are close encounters at all frequent? The answer to both parts of the question is almost always no. We recall our first course on electromagnetic theory where we met Coulomb's law and went quickly on to the notion of an *electric field* $\mathbf{E}$ and found that $\mathbf{E}$ was seldom to be found by summing the effect of each individual charge. Instead, we were introduced to the notion of a *charge density* ρ , and that $\mathbf{E}$ was to be obtained from this density, which was to be thought of as varying continuously in space. The idea of working with something like 10^{25} charges with as many calculations of $1/r^2$ was dismissed, with relief; the problem of what to do with 10^{25} singularities (at $r \rightarrow 0$) also vanished. Nearly all of the plasma physics which we will do requires knowledge only down to some scale length, with charge density (and current density) considered continuous; the finer grained behavior is omitted. Furthermore, in plasmas, with many particles in a characteristic length (which is the *Debye length*), there are relatively few close particle encounters, that is, few large angle deflections from single encounters; rather large deflections come mostly from the cumulative effect of many small deflections. Hence, we are encouraged by the nature of plasma problems to take advantage of the simplifications that come about in using a *mathematical spatial grid*, as shown in Figure 2-2b, usually fine enough to resolve a *Debye length*, in order to measure the charge density and, thence, calculate the electric field $\mathbf{E}$.

There are some exceptions to these generalities. For example, one may calculate the electric field in one-dimensional problems relatively easily without using a spatial grid. However, the grid provides a *smoothing effect* by not resolving spatial fluctuations that are smaller than the grid size; an exact field calculation would keep everything, which is usually more than we want.

The use of temporal and spatial grids, which are mathematical and not physical, causes concern about accuracy and may create what we will term *nonphysics*. However, we only touch on such in Part One, and develop this material later in Part Two. Suffice it to say: the possibility of nonphysical effects may restrict our choices of parameters on occasion, but generally these effects can be avoided; inaccuracies will always be with us and simply must be made small.

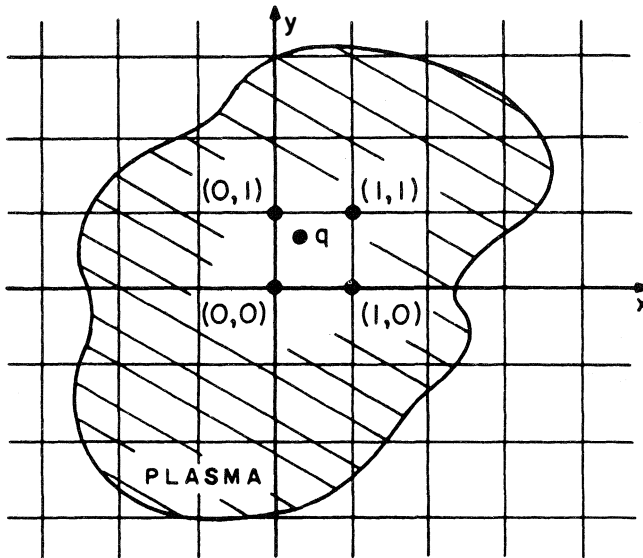


Figure 2-2b A mathematical grid is set into the plasma region in order to measure charge and current densities ρ , J ; from these we will obtain the electric and magnetic fields E , B on the grid. A charged particle q at (x,y) will typically be counted in terms of ρ at the nearby grid points $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$ and in terms of J at the faces between these points. The force on q will also be obtained from the fields at these nearby points.

PROBLEMS

2-2a Sketch the electric field $E(x)$ versus x for the 1d model of Figure 2-2a, for various boundary conditions, such as (a) potential equal to zero at $x = 0$, $x = L$, or (b) the system is periodic, with period of L , or (c) there is an applied potential difference [say, $\phi(0) = 0$, $\phi(L) = V_0$]. Let the charges have some thickness as shown in Figure 2-2a to show E within the charge. Consider neutral and nonneutral systems (equal or nonequal numbers of positive and negative charges). Consider a stationary uniform background of charge of one sign and sheet charges of another sign, with net neutrality; show $\rho(x)$, $E(x)$, and $\phi(x)$.

2-2b Let the charge density of a sheet extend from x_a to x_b , and be zero outside these values. Show that the electrostatic force on the sheet, for arbitrary charge distribution within the sheet [that is, $\rho(x)$ is arbitrary between x_a and x_b], is $q(E(x_a) + E(x_b))/2$, where q is the total charge per unit area of the charged sheet. *Hint:* Consider the integral $\int \rho E dx = \int \frac{\partial}{\partial x}(E^2/2\epsilon_0) dx$ across the sheet. This simple and ancient result is a "gift" of the one-dimensional model, one of several that we will use to advantage. (See, e.g., *Portis*, 1978, p. 68, for force on charged sheet and p. 324 for force on a current sheet.)

2-2c The text states that the fields can be found exactly analytically in one dimension so that a spatial grid is not needed solely for this purpose. Suppose that we used sheets of zero thickness and found the fields exactly at time t and then again at time $t + \Delta t$, etc.; however, these fields change only when two (or more) particles cross which may be during Δt , leading to error. What corrections would you suggest to account for such crossing(s) during Δt ? (See *Dawson*, 1970, p. 4-10, for discussion of sheet crossings.) Explain why a gridded model with finite-size particles has less of a problem in accounting for crossings.

2-3 THE COMPUTATIONAL CYCLE: GENERAL REMARKS

At each step in time, the program solves for the fields from the particles and then moves the particles; this cycle is shown in Figure 2-3a. There may be tens of steps in a characteristic period of the plasma and there may be tens of periods in a typical run, which adds up to hundreds or thousands of time steps in a given run.

The cycle starts at $t = 0$, with some appropriate initial conditions on the particle positions and velocities. The computer runs to the number of time steps it is told. Various *diagnostics* are printed out at the end of the run; some are in the form of *snapshots* at particular times, such as densities or fields or velocity distributions; some are in the form of *time histories*, such as energy versus time. These graphs are the record from which one obtains the physics of the simulation. Numbers per se are very seldom the object of a plasma simulation.

The particles are processed through the boxes shown in Figure 2-3a, much as the fields and forces are created in the actual plasma. Let us follow a cycle by starting from initial values of positions and velocities. Keep in mind that hundreds or thousands (or 10^6 and up in two and three dimensions) of particles are being processed.

The particle quantities, such as velocity and position, are known at the particle and may take on all values in $\mathbf{v}$ and $\mathbf{x}$ space, called *phase space*. The name of the particle is given by index i , such as v_i and x_i . The field quantities will be obtained only on the spatial grid, known only at discrete points in space labeled with index j such as E_j . The ties from the particle position

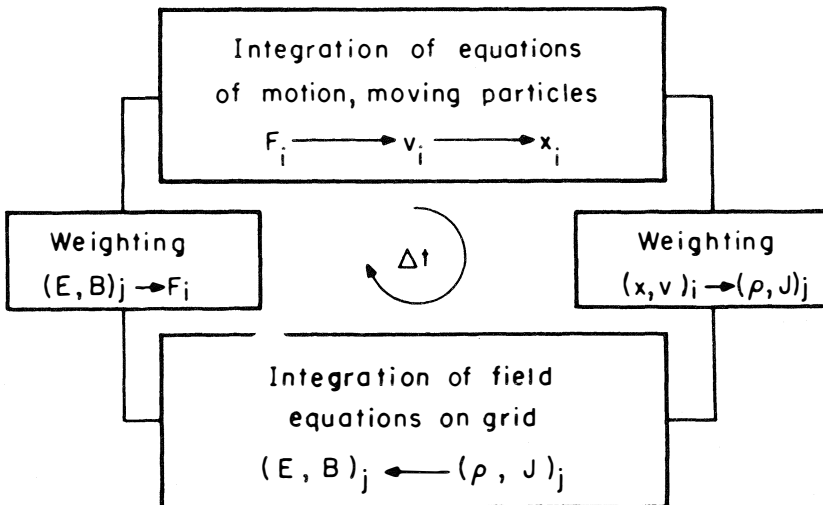


Figure 2-3a A typical cycle, one time step, in a particle simulation program. The particles are numbered $i = 1, 2, \dots, NP$; the grid indices are j , which become vectors in 2 and 3 dimensions.

and velocities to the field quantities are made by first calculating the charge and current densities on the grid; this step requires stating how to produce the grid densities from the particle positions and velocities. This process of charge and current assignment implies some *weighting* to the grid points that is dependent on particle position. Once the densities are established on the grid, then we will use various methods to obtain the electric and magnetic fields. With the fields known on the grid, but with the particles scattered around within the grid, we interpolate the fields from the grid to the particles in order to apply the force *at the particle* by again performing a weighting.

How do we distinguish particles? What information is stored for the particles and for the fields? The particles may be known by the way in which they are stored in the computer memory; they may be in some kind of order, with only their present velocity and position stored ($\mathbf{v}_i$, $\mathbf{x}_i$); their values of charge q_i and mass m_i may be put elsewhere (e.g., with only two species, electrons and ions, q and m would change only once in running all of the particles through in a time step). The fields are known at the grid points and are stored, probably indexed in an array, so that they can be recalled readily. There will almost always be *many more particles than grid points*, and external storage allows use of more particles than might fit into the computer's fast memory, along with the field quantities. Since the particles are integrated independently, only a few need be in fast memory at a time. The field quantities probably will be retained in fast memory, as in most present methods they have to be recalled randomly.

There are many variations of the cycle shown. For example, the electrons (with relatively large ω_{pe} , ω_{ce} , ω_{uh}) may be advanced at a relatively small Δt_e ; the ions (with much smaller ω_{pi} , ω_{ci} , ω_{ih}) may be advanced with a relatively large Δt_i ; the fields may be obtained on yet a third scale, Δt_f , possibly relatively short for electromagnetic fields (waves move faster than particles) or relatively large for observing low frequency effects.

2-4 INTEGRATION OF THE EQUATIONS OF MOTION

A problem may call for 10,000 particles to be run for 1000 time steps. This means that the equations of motion must be integrated 10,000 times $1000 = 10^3$ times. We want to use as fast a method as possible, and still retain acceptable accuracy. The time per particle per step currently is on the order of microseconds. Explicitly, a 10^4 particle program of 10^3 time steps running at $10 \mu\text{sec/particle/step}$ on a machine with a 720 dollar/hour charge would cost 20 dollars for the 100-second run.

In addition, our choice of method must take into account the storage capability of the computer we will use in terms of the number of quantities that may be kept for each particle. If we were to follow just the trajectory of one particle through given fields (so-called *trajectory calculations*), then we probably would choose to keep velocity and position information from

several previous time steps and use a method of time integration with a high order of accuracy. However, the minimum information needed for integration is the particle velocity and position, two words per particle or 20,000 words for our problem. Storing the v_i , x_i for several previous time steps would multiply this number. Using a high-order method (e.g., Runge-Kutta) would multiply the operations taken for each particle. Hence, we nearly always will choose to use the least information (storage) and the fastest method we can.

One commonly used integration is called the *leap-frog method*. The two first-order differential equations to be integrated separately for each particle are

$$m \frac{dv}{dt} = F \quad (1)$$

$$\frac{dx}{dt} = v \quad (2)$$

where F is the force. These equations are replaced by the *finite-difference equations*

$$m \frac{v_{\text{new}} - v_{\text{old}}}{\Delta t} = F_{\text{old}} \quad (3)$$

$$\frac{x_{\text{new}} - x_{\text{old}}}{\Delta t} = v_{\text{new}} \quad (4)$$

The flow in time and notation is shown in Figure 2-4a, which makes clear the *time centering*. The computer will advance v_t and x_t to $v_{t+\Delta t}$, $x_{t+\Delta t}$, even though v and x are not known at the same time. The user must show care in at least two ways: first, initial conditions for particle velocities and

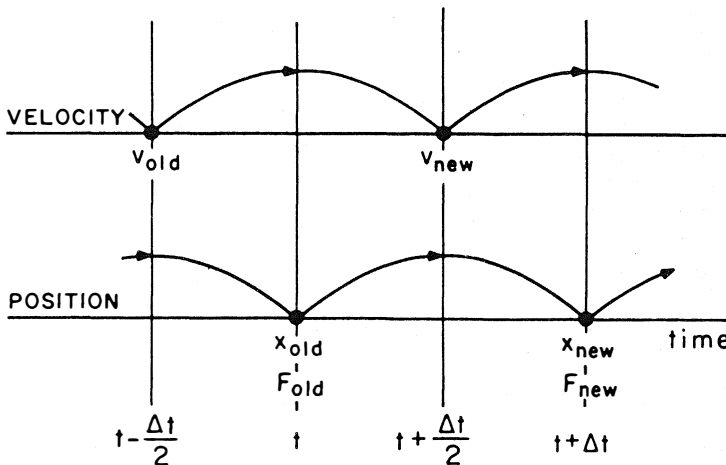


Figure 2-4a Sketch of leap-frog integration method showing time-centering of force F while advancing v , and of v while advancing x .

positions given at $t = 0$ must be changed to fit in the flow; we push $\mathbf{v}(0)$ back to $\mathbf{v}(-\Delta t/2)$ using the force $\mathbf{F}$ calculated at $t = 0$; second, the energies calculated from $\mathbf{v}$ (kinetic) and $\mathbf{x}$ (potential, or field) must be adjusted to appear at the same time.

The leap-frog method has error, with the error vanishing as $\Delta t \rightarrow 0$. We will use a leap-frog integrator in nearly all of our programs, because it is both simple (easy to understand, and with minimum storage) and surprisingly accurate (as shown later). Applying this method to integration of a simple harmonic oscillator of radian frequency ω_0 (in a later section), we will find that there is *no* amplitude error for $\omega_0 \Delta t \leq 2$ and that the phase advance for one step is given by

$$\omega_0 \Delta t + \frac{1}{24}(\omega_0 \Delta t)^3 + \text{higher-order error terms} \quad (5)$$

The error terms dictate a choice of $\omega_0 \Delta t \leq 0.3$ in order to observe oscillations or waves for some tens of cycles with acceptable accuracy.

The force $\mathbf{F}$ has two parts,

$$\mathbf{F} = \mathbf{F}_{\text{electric}} + \mathbf{F}_{\text{magnetic}} \quad (6)$$

$$= q\mathbf{E} + q(\mathbf{v} \times \mathbf{B}) \quad (7)$$

Here the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ are to be calculated *at the particle*. Hence, using a spatial grid, we must interpolate $\mathbf{E}$ and $\mathbf{B}$ from the grid to the particle; we will do this in the same way as we determined the charge density (at the grid points) from the particle positions, a point discussed in Section 2-6. As we will see later, the electric force on a particle will depend not only on the distance to other particles (physical) but also on the position within the cell (nonphysical).

For our one dimensional purpose now, let us consider the particle displacement to be along x , and that we have velocities v_x and v_y , with a uniform static magnetic field B_0 , along z . The $q(\mathbf{v} \times \mathbf{B})$ force, as shown in Figure 2-4b, is simply a *rotation* of $\mathbf{v}$; that is, $\mathbf{v}$ does not change in magnitude. However, the $q\mathbf{E} = \hat{x}qE_x$ force does alter the magnitude of $\mathbf{v}$ (v_x , that is); $E_y = 0$. Hence, a physically reasonable scheme which is centered in time is as follows (with t' and t'' as dummy variables, $t - \Delta t/2 < t' < t'' < t + \Delta t/2$):

Half acceleration

$$\begin{aligned} v_x(t') &= v_x \left(t - \frac{\Delta t}{2} \right) + \left(\frac{q}{m} \right) E_x(t) \left(\frac{\Delta t}{2} \right) \\ v_y(t') &= v_y \left(t - \frac{\Delta t}{2} \right) \end{aligned} \quad (8)$$

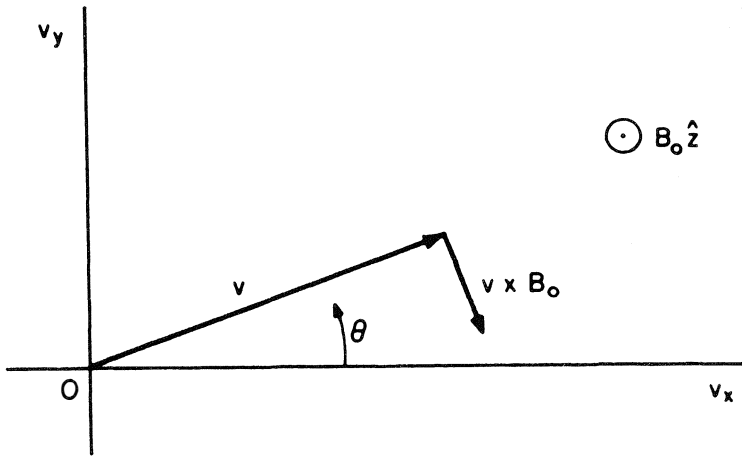


Figure 2-4b The v_x, v_y plane, showing the $q(\mathbf{v} \times \mathbf{B})$ force normal to $\mathbf{v}$, which results in a rotation of $\mathbf{v}$, with no change in speed with $\dot{\theta} < 0$ for $(q/m) > 0$, $B_0 > 0$.

Rotation

$$\begin{pmatrix} v_x(t'') \\ v_y(t'') \end{pmatrix} = \begin{pmatrix} \cos \omega_c \Delta t & \sin \omega_c \Delta t \\ -\sin \omega_c \Delta t & \cos \omega_c \Delta t \end{pmatrix} \begin{pmatrix} v_x(t') \\ v_y(t') \end{pmatrix} \quad (9)$$

Half acceleration

$$\begin{aligned} v_x \left[t + \frac{\Delta t}{2} \right] &= v_x(t'') + \left(\frac{q}{m} \right) E_x(t) \left(\frac{\Delta t}{2} \right) \\ v_y \left[t + \frac{\Delta t}{2} \right] &= v_y(t'') \end{aligned} \quad (10)$$

The angle of rotation is

$$\Delta \theta = -\omega_c \Delta t \quad \text{i.e.} \quad \dot{\theta} = -\omega_c \quad (11)$$

as desired, where the *cyclotron frequency* (radians/sec) as defined by

$$\omega_c \equiv \left(\frac{q}{m} \right) B_0 \quad (12)$$

which carries the sign of q and of B_0 . This scheme (Boris, 1970b) is elaborated upon in Chapter 4.

One complication arises at $t = 0$ when the initial conditions, $\mathbf{x}(0)$ and $\mathbf{v}(0)$, are given at the same time. The main loop runs with $\mathbf{x}$ leading $\mathbf{v}$ by $\Delta t/2$. Hence, at the start, $\mathbf{v}(0)$ is moved backward to $\mathbf{v}(-\Delta t/2)$, by first rotating $\mathbf{v}(0)$ through the angle $\Delta \theta = +\omega_c \Delta t/2$ and then applying a half acceleration using $-\Delta t/2$ (really a deceleration?) based on $\mathbf{E}(0)$ obtained from $\mathbf{x}(0)$; see Sec. 2-7 and Figure 2-7a.

2-5 INTEGRATION OF THE FIELD EQUATIONS

Starting from the charge and current densities as assigned to the grid-points, we now obtain the electric and magnetic fields, in general, from Maxwell's equations, using ρ and J as sources. Here we take this step for an electrostatic problem (meaning $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t \approx 0$ so that $\mathbf{E} = -\nabla \phi$) in one dimension x .

The differential equations to be solved are

$$\mathbf{E} = -\nabla \phi \quad \text{or} \quad E_x = -\frac{\partial \phi}{\partial x} \quad (1)$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \text{or} \quad \frac{\partial E_x}{\partial x} = -\frac{\rho}{\epsilon_0} \quad (2)$$

which are combined to obtain *Poisson's equation*

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0} \quad \text{or} \quad \frac{\partial^2 \phi}{\partial x^2} = -\frac{\rho}{\epsilon_0} \quad (3)$$

One approach is to solve the finite difference equations of (1) and (3), using the grid shown in Figure 2-5a, as

$$E_j = \frac{\phi_{j-1} - \phi_{j+1}}{2\Delta x} \quad (4)$$

$$\frac{\phi_{j-1} - 2\phi_j + \phi_{j+1}}{(\Delta x)^2} = -\frac{\rho_j}{\epsilon_0} \quad (5)$$

This last may be written in matrix form as

$$\mathbf{A}\phi = -\frac{(\Delta x)^2}{\epsilon_0}\rho \quad (6)$$

We are to use the ρ_j 's known from the x_i 's, to obtain the unknown ϕ_j 's and then the E_j 's, for j running from 0 to $L/\Delta x$ (roughly) where L is the length of the system of NG points. By using the known boundary conditions

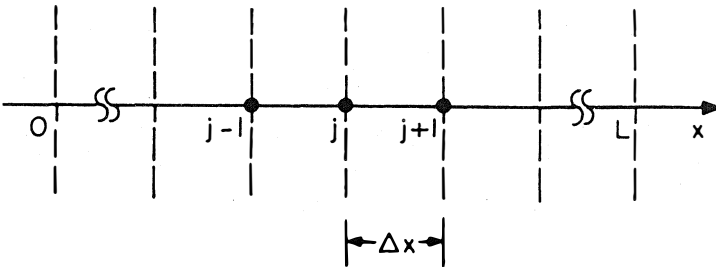


Figure 2-5a One dimensional numerical grid, with grid planes located at $X_j = j\Delta x$, uniformly spaced. The charge density ρ , the potential ϕ , and the electric field E_x will be obtained only at the X_j 's.

at $x = 0, L$ and all of the ρ_j there will be as many equations as unknowns (the ϕ); hence, the problem is solvable. There are numerous methods of solution. These are taken up in Appendix D.

A very powerful approach for *periodic systems* is to use a discrete *Fourier series* for all grid quantities. This approach also provides spatial spectral information on ρ , ϕ , and E which is useful in relating results to plasma theory, and which also allows control (smoothing) over the spectrum of field quantities. We now present this method in some detail; it is the basis for the field solver in our code ES1.

The ability to Fourier transform efficiently through use of the *fast Fourier transform FFT*, an invention of the 1960's, is a major enabling factor in this computation. The key to the solution is the assumption that, in the problems we attack, $\rho(\mathbf{x})$ and $\phi(\mathbf{x})$ have Fourier transforms, $\rho(\mathbf{k})$ and $\phi(\mathbf{k})$, where $\mathbf{k}$ is the wave vector in the Fourier transform kernel, $\exp(-i\mathbf{k}\cdot\mathbf{x})$. (We are implying certain boundary conditions by this step, such as periodic, which are considered in detail later.) This assumption allows us to obtain $\phi(\mathbf{k})$ from $\rho(\mathbf{k})$ in Poisson's differential equation directly as, in one dimension, $\partial^2/\partial x^2$ is replaced by $-k^2$; that is,

$$\phi(k) = \frac{\rho(k)}{\epsilon_0 k^2} \quad (7)$$

The next step is to take the inverse Fourier transform of $\phi(k)$ in order to obtain $\phi(x)$ and then $E(x)$ using (2). The overall sequence is given in Figure 2-5b. (The reader will quickly note that one may go directly to $E(k)$; we go into that possibility later.) The hitch in proceeding in this manner is that we have a finite Fourier series; we have $\rho(x)$ only at the X_j 's, at NG points.

The solution using a finite Fourier series starts from the charge densities at the grid points, with values $\rho(X_j)$, $j = 0, 1, 2, \dots, \text{NG} - 1$ for a total of NG values. Letting the grid functions $G(X_j)$ (standing for field or potential or charge density) be periodic, $G(X_j) = G(X_j + L)$, then the finite discrete Fourier transform is (sum on $X_j = j\Delta x$)

$$G(k) = \Delta x \sum_{j=0}^{\text{NG}-1} G(X_j) e^{-ikX_j} \quad (8)$$

The inverse transform is [the sum is on $k = n(2\pi/L)$]

$$G(X_j) = \frac{1}{L} \sum_{n=-\text{NG}/2}^{\text{NG}/2-1} G(k) e^{ikX_j} \quad (9)$$

which produces NG distinct values of $G(X_j)$. (The virtue of the FFT is that

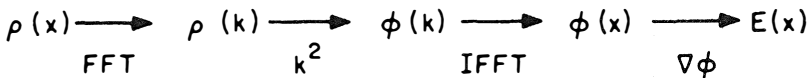


Figure 2-5b A possible sequence for solving Poisson's equation using the fast Fourier transform (FFT) and its inverse (IFFT).

it performs the sums rapidly.) Using the above series for $\rho(X_j)$, $\phi(X_j)$, and $E(X_j)$ with the particular set of finite difference equations chosen from (4) and (5), we obtain

$$E(k) = -i\kappa\phi(k) \quad (10)$$

where

$$\kappa = k \left[\frac{\sin(k\Delta x)}{k\Delta x} \right] = k \text{ dif}(k\Delta x) \quad (11)$$

where (from diffraction theory)

$$\text{dif } \theta \equiv \frac{\sin \theta}{\theta}$$

and

$$\phi(k) = \frac{\rho(k)}{\epsilon_0 K^2} \quad (12)$$

where

$$K^2 = k^2 \left[\frac{\sin \frac{k\Delta x}{2}}{\frac{k\Delta x}{2}} \right]^2 = k^2 \text{dif}^2\left(\frac{k\Delta x}{2}\right) \quad (13)$$

The finite difference terms, κ and K^2 , approach the differential equation result, k and k^2 , as the grid becomes finer, $k\Delta x \rightarrow 0$. The overall understanding of the role of the spatial grid, both in accuracy (*i.e.*, κ versus k , K^2 versus k^2) and in *aliasing* (the dynamics creates effects at $|k\Delta x| > \pi$ which are read falsely by the grid, being placed at $|k\Delta x| < \pi$) will be covered in detail in Part Two on the theory of simulation. A discussion on the difference between using k , k^2 and κ , K^2 is given in an appendix.

With the understanding that NG values of $\rho(X_j)$ are transformed to NG values of $\rho(k)$ and so on through to obtain NG values of $E(X_j)$, the solution sequence of Figure 2-5b still holds; this is commonly done by one Fourier transform, a division by K^2 , an inverse transform and, lastly, a gradient operation [by finite differencing (4)] on $\phi(X_j)$ to obtain $E(X_j)$. There are NG values of k starting from $k_{\text{minimum}} = 2\pi/L$ (L is the length of the system; see Figure 2-2a) up to $k_{\text{maximum}} = \pi/\Delta x$ and their negatives; or, in terms of wavelengths, $\lambda_{\text{maximum}} = L$ and $\lambda_{\text{minimum}} = 2\Delta x$.

PROBLEMS

2-5a $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ may also be solved for $\mathbf{E}$ by Fourier methods. In 1d, obviously $E_x(k) = \rho(k)/i k \epsilon_0$. In 3d, electrostatic still, one must add $\nabla \times \mathbf{E} = 0$. Show that these equations produce solutions

$$E(\mathbf{k}) = \frac{\mathbf{k}\rho(\mathbf{k})}{ik^2\epsilon_0}$$

where $k^2 = k_x^2 + k_y^2 + k_z^2$.

2-5b Show that (9) is the inverse of (8). Show that $G(X_{j+\text{NG}})$, as given by (9) is equal to $G(X_j)$. Show that (9) is unchanged if the sum is taken from 0 to $\text{NG} - 1$.

2-5c When $G(X_j)$ is real, there is redundancy in the NG complex values of $G(k)$ in (8). Show that $G(-k) = G^*(k)$, and that there are $\text{NG}/2 + 1$ independent values of $\text{Re } G(k)$ (cosine coefficients) and $\text{NG}/2 - 1$ independent values of $\text{Im } G(k)$ (sine coefficients).

2-5d If $G(X_j)$ and $H(X_j)$ are both real, show that cosine and sine coefficients of G and H can be extracted from the transform [using (8)] of the complex sequence $G(X_j) + iH(X_j)$. This is accomplished by calls to subroutines CPFT and RPFT2, and the inverse by calls to RPFT12 and CPFT (Appendix A).

2-5e Obtain κ in (11) and K^2 in (13) by inserting $E(X_j)$, $\phi(X_j)$, and $\rho(X_j)$ in the transform form (9) into the finite difference forms, (4), (5). Sketch (κ/k) and (κ^2/K^2) as functions of $k\Delta x$ from $k\Delta x = 0$ to π .

2-6 PARTICLE AND FORCE WEIGHTING; CONNECTION BETWEEN GRID AND PARTICLE QUANTITIES

It is necessary to calculate the charge density on the discrete grid points from the continuous particle positions and (after the fields are obtained), to calculate the force at the particles from the fields on the grid points. In Figure 2-3a, these calculations are called weighting, which implies some form of interpolation among the grid points nearest the particle. As shown later, it is desirable to use the same weighting in both density and force calculations in order to avoid a self-force (*i.e.*, a particle accelerates itself).

In *zero-order weighting*, we simply count the number of particles within distance $\pm\Delta x/2$ (one cell width) about the j^{th} grid point and assign that number [call it $N(j)$] to that point; that is, the grid density (in one dimension) is simply $n_j = N(j)/\Delta x$. This is illustrated in Figure 2-6a(a). The common name for this weighting is *nearest-grid-point* or NGP. Computationally, the counting is fast, since only one grid look-up is done. The electric field to be used in the force is that at X_j , for all particles in the j^{th} cell.

As a particle moves into the j^{th} cell (through cell boundaries at $x = X_j \pm \Delta x/2$), the grid density due to that particle jumps up; as the particle moves out ($x > X_j + \Delta x/2$ or $x < X_j - \Delta x/2$), the grid density jumps down. The density observed at j is shown in Figure 2-6a(b). We see two effects here. One is that the particle appears to have a *rectangular shape* with a width of Δx . This leads us (*i.e.*, the grid) to think that we have a collection of *finite-size particles*; hence, the physics observed will be that of such particles rather than that of point particles. Because close encounters

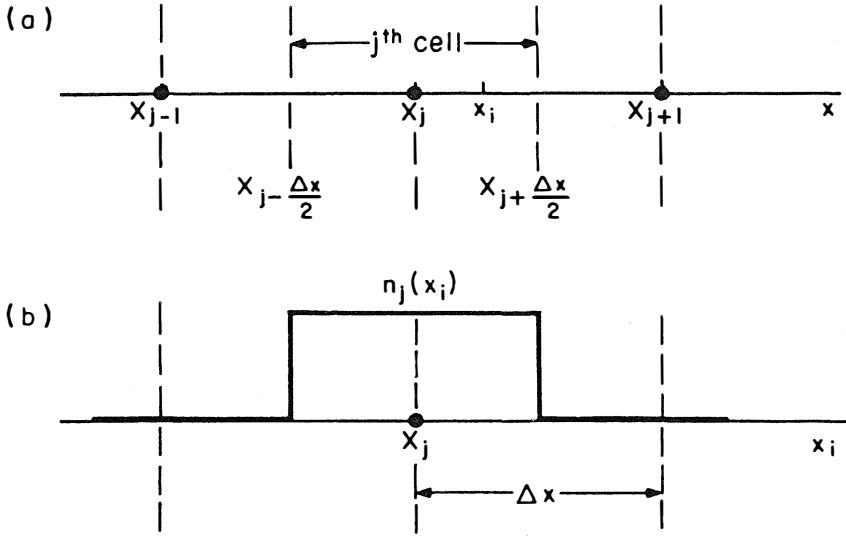


Figure 2-6a (a) Zero-order particle and field weighting, also called nearest-grid-point, or, NGP. Particles in the j^{th} cell, that is, with positions $x_i \in X_j \pm \Delta x/2$, are assigned to X_j to obtain grid density $n(X_j)$. All of these particles are acted on by the field at X_j , $E(X_j)$. (b) The density $n_j(X_j)$ at point X_j due to a particle at x_i , as the particle moves through the cell centered on X_j . This density may be interpreted as the effective particle shape.

between plasma particles are rare (*i.e.*, for many particles in a Debye length, $N_D \gg 1$, virtually all collisions are at large impact parameter), this new physics hardly alters the basic plasma effects to be studied. The second effect is that the jumps up and down as a particle passes through a cell boundary will produce a density and an electric field which are relatively *noisy* both in space and time; this noise may be intolerable in many plasma problems. Thus, we look for a better weighting.

First-order weighting smooths the density and field fluctuations, which reduces the noise (relative to zero-order weighting), but requires additional expense in accessing two grid points for each particle, twice per step. We may view this step either as an improvement in using finite-size particles or as one of better interpolation. The charged particles seem to be finite-size rigid *clouds* which may pass freely through each other. We call the model *cloud-in-cell* or CIC (*Birdsall and Fuss, 1969*). If we take the *nominal cloud* to be of uniform density and to be Δx wide as shown in Figure 2-6b(a) (the so-called square cloud), then the grid assignment is self-evident, using NGP for each element. That is, for total cloud charge of q_c , the part assigned to j is

$$q_j = q_c \left[\frac{\Delta x - (x_i - X_j)}{\Delta x} \right] = q_c \frac{X_{j+1} - x_i}{\Delta x} \quad (1)$$

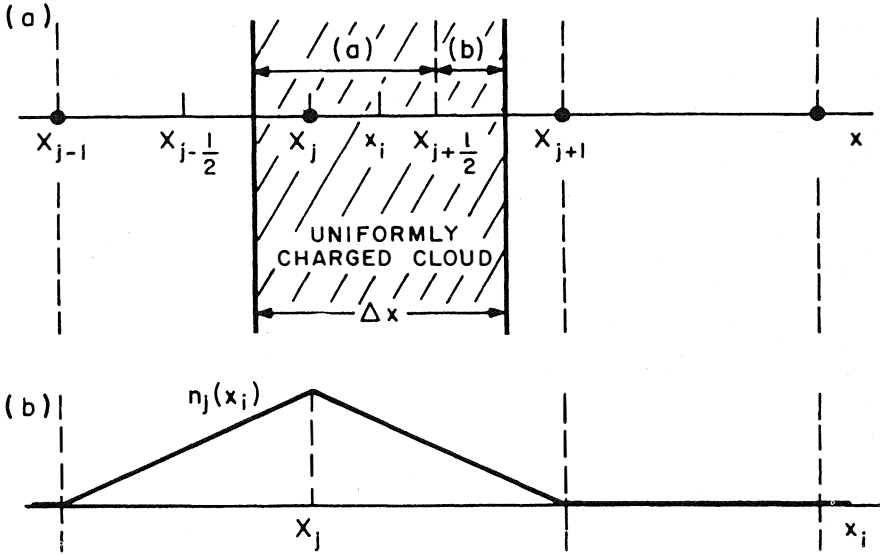


Figure 2-6b (a) First-order particle weighting, or cloud-in-cell model CIC. The nominal finite-size charged particle, or cloud, is one cell wide, with center at x_j . This weighting puts that part of the cloud which is in the j^{th} cell at X_j , fraction (a), and that part which is in the $(j+1)^{\text{th}}$ cell at X_{j+1} , fraction (b). This weighting is the same as applying NGP interpolation to each elemental part. (b) The grid density $n_j(x_i)$ at point x_i as the particle moves past X_j , again displaying the effective particle shape $S(x)$.

and the part assigned to $j+1$ is

$$q_{j+1} = q_c \left[\frac{x_i - X_j}{\Delta x} \right] \quad (2)$$

The net effect is to produce a *triangular* particle shape $S(x)$ which has width $2\Delta x$. In computation, the nearest left-hand grid point j is located first, so that $x_i > X_j$ always; then the weights are calculated and the charges assigned. Note that assignment of a point charge at x_i to its nearest grid points by linear interpolation would produce the same result; this viewpoint is called *particle-in-cell*, or PIC modeling. As a cloud moves through the grid, it contributes to density much more smoothly than with zero-order weight, as seen from Figure 2-6b(b); hence, the resultant plasma density and field will have much less noise and be more acceptable for most plasma simulation problems.

Higher-order weighting by use of quadratic and cubic *splines* rounds off further the roughness in particle shape and reduces density and field noise, but at the cost of more computation. The use of splines for higher-order weighting is discussed later. Also, the effective particle shape may be altered during the field calculation after the charge density $\rho(x)$ is Fourier transformed to $\rho(k)$, e.g., by cutting off $\rho(k)$ at some k_{last} or multiplying

$\rho(k)$ by, say, $\exp [-(k/k_{\text{last}})^n]$.

The *field or force weighting* operates in the same manner. The NGP force comes from the field at the nearest grid point. The *first-order force* (CIC-PIC) comes from linear interpolation, exactly as in the charge assignment; for a particle at x_i ,

$$E(x_i) = \left[\frac{X_{j+1} - x_i}{\Delta x} \right] E_j + \left[\frac{x_i - X_j}{\Delta x} \right] E_{j+1} \quad (3)$$

The first-order weighting consumes more computer time per particle than does the zero-order weighting; however, for a given noise level, CIC allows both a coarser grid and fewer particles than NGP, and thus regains some of the additional computer time required per particle. We will also see that higher-order weighting and smoothing tend to reduce nonphysical effects.

PROBLEM

2-6a Show that second-order Lagrangian interpolation results in discontinuities. (Splines employ a piecewise-polynomial representation also, but the segments are joined smoothly.)

2-7 CHOICE OF INITIAL VALUES; GENERAL REMARKS

We have presented some ideas on how the computational cycle functions. Now we will say something about initiating the program. Let us suppose that the problem of interest has been looked at thoroughly so that we have done the design work of choosing:

- (a) The numbers of particles and grid cells
- (b) The weighting and smoothing
- (c) The desired initial distribution function, $f(\mathbf{x}, \mathbf{v}, t = 0)$ including the initial perturbation (if any, random or ordered)

The next step is to place the particles in phase space $(\mathbf{x}, \mathbf{v})$ so that the problem desired is properly set up to run.

A cold, uniform periodic plasma of mobile electrons and immobile ions ($m_i/m_e \rightarrow \infty$) is simplest. The electrons are put in uniformly, one or more per cell. The charge density and or field solver automatically puts in a uniform neutralizing ion background by having the $k = 0$ component of ρ and E vanish. The plasma wave may be excited by perturbing the uniform positions x_{i0} by

$$x_i(t = 0) = x_{i0} + x_{i1} \cos(k_s x_{i0})$$

where $k_{\min} \leq k_s \leq k_{\max}$ is some wave vector for which we want the plasma

behavior. These x 's and the appropriate velocities (here, all zero) are put into the $t = 0$ step. The program then finds the fields at $t = 0$ from which the velocities at $t = -\Delta t/2$ are found (only used once in a run) (Section 2-4 and Figure 2-7a). Then the cycle proceeds forward by advancing $v(-\Delta t/2)$ to $v(+\Delta t/2)$, then $x(0)$ to $x(\Delta t)$, and so on for as many steps as desired.

Changing to a warm plasma requires that each particle be given a velocity v such that over some specified region (perhaps several cells, the shortest λ kept), the desired velocity distribution of velocities is well approximated. Suppose that the desired velocity distribution is flat, from $-v_{\max}$ to $+v_{\max}$ as shown in Figure 2-7b(a). Then, where the cold model had, say, four particles per cell with zero velocity, we now may split each one (figuratively) into four parts, placed for example, as shown in Figure 2-7b(b). The overall results, cold and warm, are shown in Figure 2-7c. One can improve on these placements and usually must do so [e.g., to avoid the potential multibeam instability of the approximation in Figure 2-7b and Figure 2-7c to $f(v)$ of Figure 2-7b(a)]. Such improvements will be taken up later, as well as prescriptions for loading less-simple $f(x, v)$ such as Gaussian in v_i (a Maxwellian) and nonuniform in x_i are taken up in Chapter 16.

PROBLEMS

2-7a Show that the initial half-step backward has the same-order accuracy as the main program mover, that is, to order $(\Delta t)^2$.

2-7b Provide a sketch showing how the half-step sequence can be used after the main program has been started in order to obtain x and v at the same time; i.e., go from $x(t)$ and $v(t - \Delta t/2)$ to $x(t)$ and $v(t)$. Logic says half acceleration and then half rotation will work. Is this correct? With $x(t)$ and $v(t)$, the code may then be re-started, say, to run backward (as a check, by changing the sign of Δt) or to run forward with a new Δt .

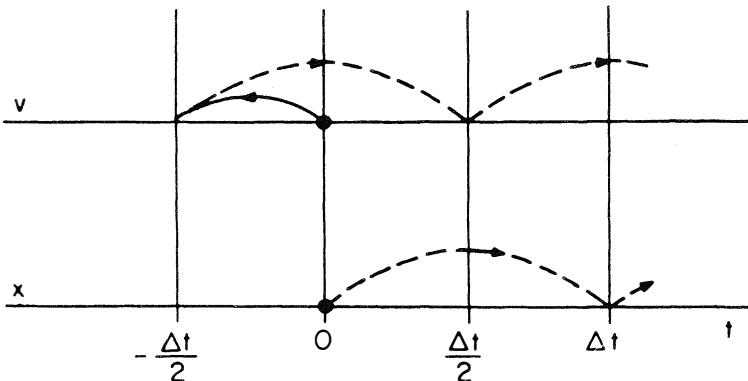


Figure 2-7a At $t = 0$, the x_i 's ($t = 0$) and v_i 's ($t = 0$) are put in, as shown. The very first step is to calculate the fields at $t = 0$ from the x_i 's and move the v_i 's back a half step, as shown by the solid line. Then the program advances the v_i 's then the x_i 's as shown earlier in Figure 2-4a.

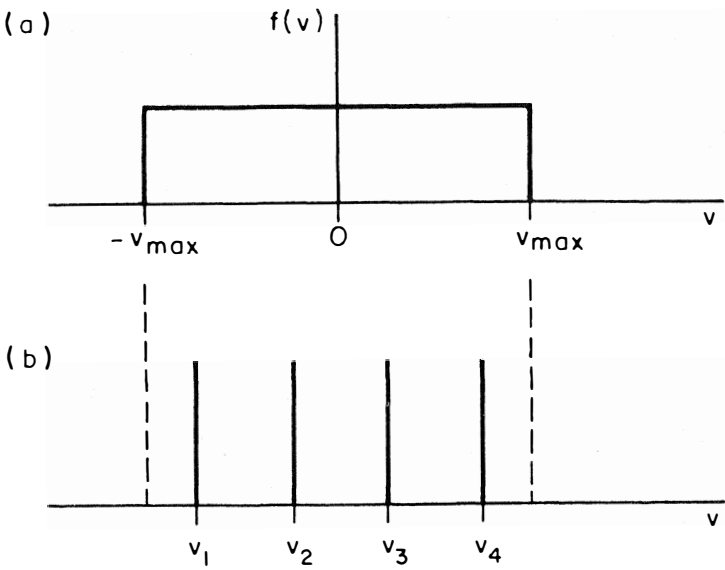


Figure 2-7b (a) The desired velocity distribution $f(v)$. (b) An elementary way of approximating $f(v)$ with four velocities.

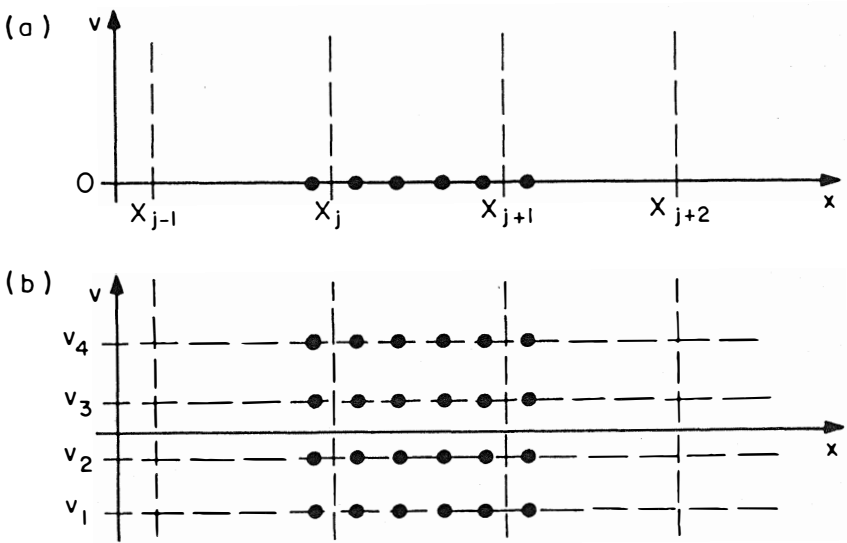


Figure 2-7c (a) Placement of particles in particle phase space for a cold uniform plasma, four particles per cell. (b) Placement of particles for a warm plasma with flat $f(v)$, using four velocities, making four beams.